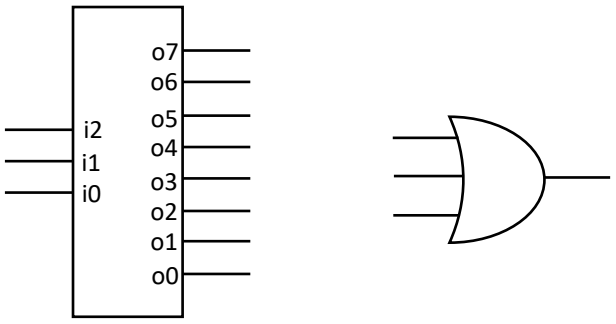
6/12/2016

EE 241 DIGITAL CIRCUITS
MIDTERM 2 Solution Manual

Name:

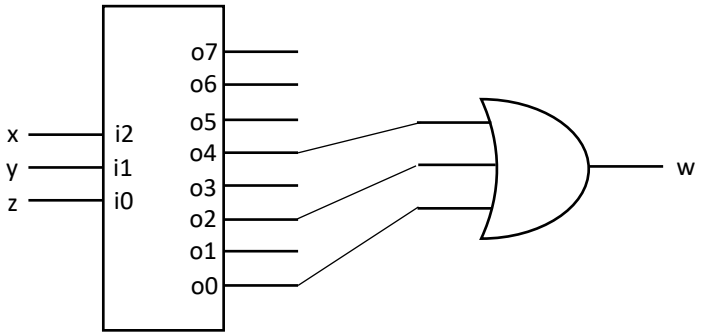
Q1) (20 points) A combinational circuit is represented by the truth table below. Implement this circuit using **one decoder** and **one three-input OR gate**.

Inputs			Output
x	y	z	w
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

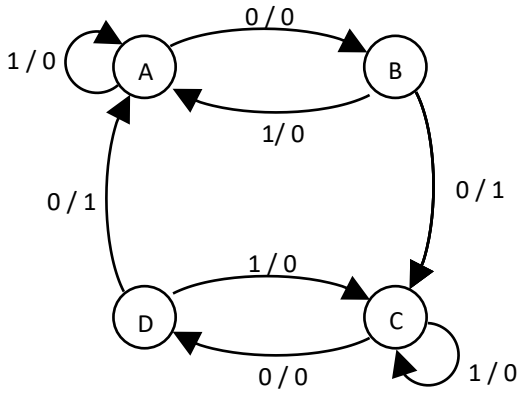


Note: Verilog description is NOT required for this question.

Solution:



Q2) (30 points) A sequential circuit is represented by the state diagram below.



Input to the circuit is x. Output of the circuit is y. Implement this sequential circuit in Verilog using case statements.

- Constructing state transitions will be awarded by 20 points in the Verilog description.
- Constructing the output will be awarded by 10 points in the Verilog description.
- Besides, no partial points will be granted in this question.

Solution:

```
module question2(y,state,x,clk,clr);

input x,clk,clr;
output reg y;

output reg [1:0] state=2'b00;

//reg [1:0] state=2'b00;
parameter A=2'b00, B=2'b01, C=2'b10, D=2'b11;

always @ (posedge clk)
begin
  if (clr == 1) state <= A;
  else
  begin
    if (x == 0)
      case (state)
      A:  state <= B;
      B:  state <= C;
      C:  state <= D;
      D:  state <= A;
      endcase
    else
      case (state)
      A:  state <= A;
      B:  state <= A;
      C:  state <= C;
      D:  state <= C;
      endcase
    if ((x == 0) && ((state == B) |(state == D))) y <= 1;
```

```
      else y <= 0;
   end
end
endmodule
```

testbench file

```
`timescale 1ns / 1ps
module question2_tb;

// Inputs
reg x,clk,clr;

// Output
wire y;
wire [1:0] state;


// Instantiate the Unit Under Test (UUT)
question2 uut (y,state,x,clk,clr);

initial begin
#2 clk=0;
forever
#2 clk=~clk;
end

initial
begin
x = 1'b0;
clr = 1'b0;

#1 x = 1'b0;
#5 x = 1'b0;
#5 x = 1'b0;
#5 x = 1'b0;
#5 x = 1'b1;
#5 x = 1'b0;
#5 x = 1'b1;

#5 clr = 1'b1;
#5 clr = 1'b0;
end
endmodule
```

Q3) (50 points) Using a frequency divider and a counter, we will blink the LED led[0] on the Basys3 board every second. In other saying, the rightmost LED on the board will turn on one second and turn off one second periodically.

a- (20 points) Construct a synchronous frequency divider module in Verilog. Input to the module will be the clock of Basys3 board which has frequency 100 MHz. Output of the module will be another clock with frequency 100/8 MHz.

Solution:

```
//clock divider
always @ (negedge clk)
begin
Fout <= Fout + 1'b1;
clk1 <=Fout[2];
end
```

b- (30 points) Write a complete top module in Verilog to turn on and off the LED (led[0]) every second. To do so, construct a 24-bit counter. To note here, 2^{24}=16,777,216. Therefore, the counter can count from 0 to 16,777,215 in decimal. Use the Verilog description for frequency division in part a.

Hints: I. $period = \frac{1}{frequency}$ II. $\frac{8}{100} * 10^{-6} * 12.5 * 10^{6} = 1$

Solution:

```
module Top_Module(led,clk);

input clk;
output reg [0:0] led;

reg [2:0] Fout;
reg clk1;

reg [23:0] count;

initial
begin
Fout=3'b000;
count=24'h000000;
end;

//clock divider
always @ (negedge clk)
begin
Fout <= Fout + 1'b1;
clk1 <=Fout[2];
end

//blink the LED
always @ (posedge clk1)
begin
count <= count + 1'b1;
if (count ==24'd1250000) led[0]<=~led[0];
end

endmodule
```